

DSM — Question Paper Topic Mapping

Dynamic Systems Modeling (702MH0C025) · B.Tech Mechatronics Sem IV · AY 2025-26

All three examination sets compared question-by-question · Prepared by Dr. Sunny Nanade, Course Instructor

Set A

Set B

Set C

Note: Q2–Q7 → Solve any FOUR (4×20 = 80 marks) · Q1 is Compulsory

Q1 — Compulsory (Answer ALL 4 parts · 5 marks each = 20 marks)

Q. No.	SET A	SET B	SET C
Q1(a) [5M]	Rigid Body Definitions (DOF, Rolling-without-slipping, Instantaneous Centre of Rotation)	Holonomic vs Non-holonomic Constraints & DOF Formula (with examples)	Three Damping Regimes (Underdamped / Critically-damped / Overdamped; automotive context)
Q1(b) [5M]	Projectile Motion — Numerical (max height, time of flight, range, velocity at $t=1$ s)	2D Kinematics — Numerical (position, velocity, acceleration from parametric $x(t)$, $y(t)$)	Rocket Kinematics — Numerical ($v(t)$, $a(t)$ from height function, max height & time)
Q1(c) [5M]	Atwood Machine — Numerical ($m_1=5$ kg, $m_2=3$ kg; acceleration, tension, velocity @ 1.5 s)	Block on 40° Rough Incline (normal force, friction force, downward acceleration)	Atwood Machine — Numerical ($m_1=6$ kg, $m_2=4$ kg; acceleration, tension, velocity @ 1 s)
Q1(d) [5M]	Work-Energy Theorem on Incline (rough 35° incline, block from rest, speed after 4 m travel)	Rotating Disk — Angular Dynamics (MOI, angular velocity, tangential & centripetal acceleration at rim)	Coordinate Conversions (Cartesian→Polar 2D; Cylindrical→Cartesian 3D)

Q2 — Pure Numerical: Particle Kinetics / Constrained Motion (20 marks)

Q. No.	SET A	SET B	SET C
Q2(a) [5M]	Frictionless Incline Block (acceleration, stopping distance, time to stop; 30°, $v_i=10$ m/s)	Vertical Circular Loop — $R=8$ m (min entry speed to complete loop; normal force at top, $m=2$ kg)	Rough Incline Block — Up & Slide (deceleration going up, check slide-back, rebound acceleration; 30°, $\mu_k=0.2$)
Q2(b) [5M]	Block-and-Tackle Pulley — 4 pulleys (MA, system efficiency @95% each, actual force, heat wasted)	Block-and-Tackle Pulley — 6 pulleys (MA, system efficiency @92% each, actual force, heat wasted; 2000 kg)	Atwood-type Elevator System (loaded car vs counterweight; acceleration, cable tension)
Q2(c) [5M]	Vertical Circular Loop — $R=2$ m (min speed at top, min entry speed; normal force @ bottom, $v=12$ m/s)	Atwood Machine — $m_1=7$ kg, $m_2=5$ kg (acceleration, tension, velocity @ 2 s; verify T range)	Frictionless 30° Incline from Rest (normal force, acceleration, velocity & distance at $t=3$ s)
Q2(d) [5M]	Movable Pulley System (constant-velocity force; acceleration for $F=300$ N; 50 kg load)	CNC 3-Axis Linear Interpolation (travel distance, unit vector, axis speeds, move time)	CNC Trapezoidal & Triangular Profiles (two move distances: 250 mm & 10 mm; times and peak velocities)

Q3 — Pure Numerical: Rigid Body Dynamics & Vibrations (20 marks)

Q. No.	SET A	SET B	SET C
Q3(a) [5M]	Rolling Race on 30° Incline (solid sphere vs disk vs hollow cylinder; acceleration, time, velocity; winner?)	Logarithmic Decrement (from successive amplitude peaks; ζ , ω_n , f_n , spring stiffness)	Solid Disk Rolling on 20° Incline (acceleration, velocity, time; compare with frictionless sliding)

Q. No.	SET A	SET B	SET C
Q3(b) [5M]	Physical Pendulum (uniform rod pivoted at end; MOI, natural frequency, period; compare with simple pendulum)	Rolling Wheel — Acceleration Analysis (ω , α from center kinematics; total acceleration at top & bottom)	Simple Pendulum Released from 60° (height gained, speed at bottom; tension in string at lowest point)
Q3(c) [5M]	Undamped Spring-Mass System (ω_n , period, max velocity, total mechanical energy)	Rotating Disk with Viscous Friction (MOI, time constant, steady-state angular velocity)	Car Braking — Energy Method (KE from 100 km/h; stopping distance, heat dissipated)
Q3(d) [5M]	Spring-Mass-Damper — Classification (ω_n , c_{cr} , ζ , ω_d , system type; time to decay to 10% amplitude)	Two-Stage Gear Train (overall gear ratio, output RPM, output torque, power balance)	Damped Vibration Parameters (ω_n , ω_d , c_{cr} , c ; cycles to halve amplitude)

Q4 — Theory (10M) + Numerical (10M): Kinematics / Rotating Frames (20 marks)

Q. No.	SET A	SET B	SET C
Q4(a) [5M]	Transport Theorem — Derivation (vector in fixed vs rotating frame; velocity relation, physical significance)	Rigid Body Velocity Formula — Derivation ($\mathbf{v}_P = \mathbf{v}_{cm} + \omega \times \mathbf{r}$; rolling constraint derivation)	Moment of Inertia & Parallel Axis Theorem (definition, PAT statement; derive I for rod about centre & end)
Q4(b) [5M]	Acceleration in Rotating Frame (derive full equation; Coriolis, centripetal, Euler terms explained)	Rotation Matrices & Gimbal Lock (R_x , R_y , R_z matrices; gimbal lock explained; quaternion fix)	Slider-Crank Geometry — Theory (constraint equation, piston position; asymmetric motion & effect of L/r)
Q4(c) [5M]	Rotating Platform — Full Numerical (Coriolis, centripetal, relative; total acceleration; $\Omega=3$ rad/s, $r=1$ m)	Rolling Wheel — Velocity Analysis (velocities at top, bottom, right-side contact point; $R=0.5$ m)	MOI Calculations — Multiple Shapes (disk, rod about centre/end, sphere; verify PAT for rod)
Q4(d) [5M]	Coriolis Deflection on Earth (latitude 45°N , $v=500$ m/s north; a_{Coriolis} , lateral deflection @ 10 s)	Rotating Rod — Kinematics ($\omega=5$ rad/s, $\alpha=2$ rad/s ² ; velocities & accelerations at 3 positions)	Slider-Crank at 6000 RPM, $\theta=90^\circ$ (ω , ϕ , piston velocity; check if acceleration > 500 g at TDC)

Q5 — Theory (10M) + Numerical (10M): Energy Methods / Lagrangian (20 marks)

Q. No.	SET A	SET B	SET C
Q5(a) [5M]	Euler-Lagrange Equation — Derivation (from Hamilton's principle; generalized coordinates, advantage)	Work-Energy Theorem (statement, conservative vs non-conservative forces, modified form)	Simple Pendulum via Lagrangian (derive EOM using $L = T - V$; Euler-Lagrange applied)
Q5(b) [5M]	Newton vs Lagrange — Comparison (force-based vs energy-based; advantages, limitations, pendulum example)	Impulse-Momentum Theory ($J = \Delta p$, coefficient of restitution, conservation conditions)	Normal Modes & Modal Analysis (eigenvalue problem in vibrations; ω_n & mode shapes from $[M]$ & $[K]$)
Q5(c) [5M]	Frictionless Ramp + Rough Surface (energy conservation to find v_{bottom} ; work-energy to find stopping distance)	Car Acceleration — Power & Energy (1200 kg, 20→40 m/s in 8 s; engine force, average & instantaneous power)	Simple Pendulum Numerical ($L=1.5$ m, $m=2$ kg; Lagrangian, EOM, period; compare with physical pendulum)
Q5(d) [5M]	Ball-Bat Impulse Problem (impulse, average force; KE before/after; energy NOT conserved)	Partially Elastic Collision — Numerical ($e=0.5$; velocities post-collision; impulse, KE loss)	Numerical ODE Solvers — RK4/Euler (Forward Euler, RK2, RK4; accuracy order, stability, why RK4)

Q6 — FBD (10M) + Difficult Numerical (10M) (20 marks)

Q. No.	SET A	SET B	SET C
Q6(a) [10M] (FBD)	Conveyor Belt Transfer — FBD (2 kg package, $v=1.5 \rightarrow v=3.0$ m/s; friction, acceleration, heat dissipated)	Mobile Manipulator on AGV — FBD ($v_{AGV}=0.5$ m/s, arm $L=1.2$ m, $\theta=30^\circ$; tip velocity, vibration isolator design)	Rotating Sensor Array — FBD ($\omega=2$ rad/s, sensor at $r=0.5$ m; centripetal & Coriolis accels, constraint forces)
Q6(b) [10M] (Numerical)	Slider-Crank Mechanism ($r=50$ mm, $L=150$ mm, 3000 RPM, $\theta=30^\circ$; ϕ , piston position, piston velocity)	CNC Trapezoidal Velocity Profile (200 mm & 15 mm moves; acceleration/deceleration times, total time)	Rough Incline + Horizontal Surface (3 kg, 35° , $\mu_k=0.25$ on incline; $\mu_k=0.3$ on flat; total friction heat)

Q7 — FBD (10M) + Difficult Numerical (10M) (20 marks)

Q. No.	SET A	SET B	SET C
Q7(a) [10M] (FBD)	Yo-Yo Dynamics — FBD ($m=50$ g, $R=30$ mm, $r=10$ mm; acceleration, tension, angular acc., % rotational KE)	Spring-Mass-Damper — Full Analysis (FBD + EOM classification; free response $x(t)$ with initial conditions)	Conveyor Belt Transfer — FBD (2 kg package, dual-belt transfer; friction forces, time, heat dissipated)
Q7(b) [10M] (Numerical)	2-DOF Mass-Spring System ($[M]$ & $[K]$ matrices; eigenvalue equation; two natural frequencies & mode shapes)	Double Pendulum (T & V expressions; Euler-Lagrange EOMs; small-angle linearisation; chaos comment)	Compound Gear Train (2 stages; overall gear ratio; output speed, torque, power verification)

QBMS Format = Final QP content (identical questions, different layout). Synoptic Table is identical across all 3 sets — same CO/BL/Unit coverage. Dr. Sunny Nanade, Course Instructor · NMIMS STME Indore